

RAG System Evaluation Report

Breast Cancer Screening Knowledge Base — Query & Retrieval Review

Total queries reviewed	4
Source documents	USPSTF Breast Cancer Screening: Final Recommendation Statement; Evidence Re
Retrieval tag observed	CLINICAL

Legend: ■ Strong / mostly relevant ■ Partially relevant ■ Weak retrieval

Contents

1. What age should women start breast cancer screening? — **Strong retrieval**
2. How often should women get mammography? — **Partially relevant**
3. What are the recommendations for women aged 75 years or older? — **Mostly relevant**
4. Is supplemental MRI recommended for women with dense breasts? — **Weak retrieval**

Query 1 What age should women start breast cancer screening?

Retrieved chunk 1 | Tag: **CLINICAL** | Source: breast-cancer-screening-final-rec - Copy.pdf | ID: breast-cancer-screening-final-rec - Copy.pdf_35

The most recent epidemiologic data reviewed by the USPSTF show greater incidence of breast cancer at younger ages, and decision modeling shows a greater magnitude of benefit for screening women between the ages of 40 and 49 years. The USPSTF considered both these lines of evidence as it issued its current B recommendation for biennial screening mammography for women aged 40 to 74 years. Last, in response to comments, the USPSTF added the breast cancer screening recommendations from the American College of Radiology to the Recommendations of Others section. Research Needs and Gaps: See Table 3 for research needs and gaps related to screening for breast cancer. Recommendations of Others: The American Cancer Society recommends that women with an average risk of breast cancer should undergo regular screening mammography starting at age 45 years. It suggests that women aged 45 to 54 years should be screened annually, that women 55 years or older should transition to biennial screening or have the opportunity to continue screening annually, that women should have the opportunity to begin annual screening between the ages of 40 and 44 years, and that women should continue screening mammography as long as their overall health is good and they have a life expectancy of 10 years or longer. The American College of Obstetricians and Gynecologists recommends that women at average risk of breast cancer should be offered screening mammography starting at age 40 years, using shared decision-making, and if they have not initiated screening in their 40s, they should begin screening mammography by no later than age 50 years.

Retrieved chunk 2 | Tag: **CLINICAL** | Source: breast-cancer-screening-final-rec - Copy.pdf | ID: breast-cancer-screening-final-rec - Copy.pdf_36

Beyond age 75 years, the decision to discontinue screening mammography should be based on shared decision-making informed by the woman's health status and longevity. The American College of Radiology and the Society of Breast Imaging recommend annual screening mammography beginning at age 40 years for women at average risk. They recommend that screening should continue past age 74 years, without an upper age limit, unless severe comorbidities limit life expectancy. The American College of Radiology also recommends breast cancer risk assessment by age 25 years for all individuals. The American Academy of Family Physicians supports the 2016 USPSTF recommendation on screening for breast cancer.

Evaluator comment

Strong retrieval

Both chunks directly answer the question. They lay out the USPSTF's B-grade recommendation (biennial screening, ages 40-74) alongside the differing start-age guidance from ACS (45, with option from 40), ACOG (40, shared decision-making), and ACR/SBI (40, annual). This is exactly the comparative, guideline-level content a clinician would need.

Query 2 How often should women get mammography?

Retrieved chunk 1 | Tag: **CLINICAL** | Source: breast-cancer-screening-final-rec - Copy.pdf | ID: breast-cancer-screening-final-rec - Copy.pdf_35

(Same chunk as Query 1, ID _35 -- see above.) Covers USPSTF's biennial recommendation for ages 40-74, plus ACS annual/biennial guidance by age band and ACOG's every-1-or-2-years recommendation.

Retrieved chunk 2 | Tag: **CLINICAL** | Source: breast-cancer-screening-final-evidence-review - Copy.pdf | ID: breast-cancer-screening-final-evidence-review - Copy.pdf_385

The analytic sample included 7,602,869 screening mammograms conducted among 4,580,698 women. Know Your Risk: Assessment at Screening (KYRAS) study: The KYRAS study calculated the cumulative risk of false-positive screens over a median of 8.9 years. Eligible women were those screened at the Columbia University Medical Center (New York) during 2014 and 2015; for these women, the study collected information on previous mammograms going back to 1989 based on their health record (N = 2,019; median age, 59 years). Women with a previous diagnosis of breast cancer or who did not speak English or Spanish were excluded from the study. Frequency of screening was determined in the electronic health record by calculating the median number of days between mammograms. If the median screening interval was between 274 days (9 months) and 548 days (18 months), then it was coded as yearly screening; a median interval between 548 days (18 months) and 913 days (30 months) was coded as biennial screening. Overall, women underwent a median of 7 mammograms during the study period.

Retrieved chunk 3 | Tag: **CLINICAL** | Source: breast-cancer-screening-final-evidence-review - Copy.pdf | ID: breast-cancer-screening-final-evidence-review - Copy.pdf_290

False-positive mammography rates were highest among women ages 40 to 49 years (121.2 per 1,000 women [95% CI, 105.6 to 138.7]) and declined with age. False-negative rates were low across all age groups (ages 40 to 49 years; 1.0 per 1,000 women [95% CI, 0.9 to 1.2]). Rates of recommendations for additional diagnostic imaging were highest among women ages 40 to 49 years (124.9 per 1,000 women [95% CI, 109.3 to 142.3]) and decreased with age, while rates of recommendations for biopsy did not differ between age groups (ages 40 to 49 years; 16.4 per 1,000 women [95% CI, 13.2 to 20.3]). Rates of invasive breast cancer were lowest among women ages 40 to 49 years (2.2 per 1,000 women [95% CI, 1.8 to 2.6]) and increased across age groups (P<0.001).

Evaluator comment

Partially relevant

Only the first chunk (reused from Query 1) actually answers the frequency question, contrasting USPSTF's biennial recommendation with ACS/ACOG annual-to-biennial guidance. The second chunk (KYRAS study) describes how screening interval was measured in a cohort study rather than a recommendation, and the third chunk reports false-positive/negative rates by age -- relevant to screening trade-offs but not to "how often." The retriever appears to have keyed on "mammography"/"screening interval" surface terms rather than the intent of the question; a rerank step or a query rewrite toward "screening interval recommendation" would likely improve precision here.

Query 3

What are the recommendations for women aged 75 years or older?

Retrieved chunk 1 | Tag: **CLINICAL** | Source: breast-cancer-screening-final-rec - Copy.pdf | ID: breast-cancer-screening-final-rec - Copy.pdf_31

Response to Public Comment: A draft version of this recommendation statement was posted for public comment on the USPSTF website from May 9, 2023, to June 6, 2023. The USPSTF received many comments on the draft recommendation and appreciates all the thoughtful views and perspectives that were shared. Many comments agreed with the draft recommendation. Several comments suggested that there should be no upper age limit for breast cancer screening or that an upper age should be based on life expectancy. In response, the USPSTF notes that no trials of breast cancer screening enrolled women 75 years or older and an emulated trial showed no benefit to screening women aged 75 to 79 or 80 to 84. Some comments suggested that breast cancer screening should start prior to age 40 years... Several comments expressed that breast cancer screening should be recommended annually. In response, the USPSTF would like to reiterate that it did not identify any randomized trials directly comparing annual vs biennial screening.

Retrieved chunk 2 | Tag: **CLINICAL** | Source: breast-cancer-screening-final-evidence-review - Copy.pdf | ID: breast-cancer-screening-final-evidence-review - Copy.pdf_32

Women in these high-risk groups require different screening regimens than the general population. Non-modifiable factors associated with increased risk among women eligible for routine screening include increased age; at least one first- or second-degree relative with breast cancer; and heterogeneously or extremely dense breast tissue. Women in their 40s with extremely dense breast tissue or at least one first-degree relative with breast cancer are estimated to have a twofold increased risk of breast cancer. Additional risk factors that have been associated with an increased risk of breast cancer include use of menopausal hormone therapy, increasing body mass index (BMI), alcohol use, and nulliparity or giving birth after age 35. Risk factors associated with a decreased risk of breast cancer include breastfeeding and increased physical activity.

Retrieved chunk 3 | Tag: **CLINICAL** | Source: breast-cancer-screening-final-rec - Copy.pdf | ID: breast-cancer-screening-final-rec - Copy.pdf_36

(Same chunk as Query 1, ID _36 -- see above.) States that the decision to discontinue screening past age 75 should rest on shared decision-making, and that ACR/SBI recommend continuing annual screening past age 74 without an upper limit unless severe comorbidities limit life expectancy.

Evaluator comment

Mostly relevant

The first and third chunks are on-target: they explain that no screening trials enrolled women 75 or older, that an emulated trial showed no benefit for ages 75-79 or 80-84, and that ACR/SBI recommend continuing past 74 without an upper limit absent severe comorbidities -- directly answering the question with the appropriate nuance (evidence gap + guideline split). The second chunk, on non-modifiable risk factors like breast density and family history, is topically adjacent (breast cancer risk) but doesn't address the 75+ age question and dilutes the result set.

Query 4

Is supplemental MRI recommended for women with dense breasts?

Retrieved chunk 1 | Tag: **CLINICAL** | Source: breast-cancer-screening-final-evidence-review - Copy.pdf | ID: breast-cancer-screening-final-evidence-review - Copy.pdf_312

Radiology. 2021;299(2):278-286. a. Bakker MF, de Lange SV, Pijnappel RM, et al. Supplemental MRI Screening for Women with Extremely Dense Breast Tissue. New England Journal of Medicine. 2019;381(22):2091-102. b. de Lange SV, Bakker MF, Monninkhof EM, et al. Reasons for (non)participation in supplemental population-based MRI breast screening for women with extremely dense breasts. Clinical Radiology. 2018;73(8):759.e1-e9. Appendix D. Excluded Studies -- Breast Cancer Screening, Kaiser Permanente Research Affiliates EPC.

Retrieved chunk 2 | Tag: **CLINICAL** | Source: breast-cancer-screening-final-evidence-review - Copy.pdf | ID: breast-cancer-screening-final-evidence-review - Copy.pdf_39

In addition to primary screening, the value of supplemental screening for women with dense breasts is an active area of research. The lack of a standardized and reliable assessment tool for measuring breast density, and its variability over a woman's lifespan, pose challenges for research into the optimal screening strategy for women identified as having dense breasts. The supplemental screening modalities used to screen women with dense breasts include handheld breast ultrasonography, automated whole breast ultrasound, magnetic resonance imaging (MRI), and DBT. Limited data are available for evaluating its performance in average-risk populations with only breast density as a risk factor.

Retrieved chunk 3 | Tag: **CLINICAL** | Source: breast-cancer-screening-final-rec - Copy.pdf | ID: breast-cancer-screening-final-rec - Copy.pdf_52

Reference list, items 68-74, including: Bakker MF, de Lange SV, Pijnappel RM, et al; DENSE Trial Study Group. Supplemental MRI screening for women with extremely dense breast tissue. N Engl J Med. 2019;381(22):2091-2102. Veenhuizen SGA, de Lange SV, Bakker MF, et al; DENSE Trial Study Group. Supplemental breast MRI for women with extremely dense breasts: results of the second screening round of the DENSE trial. Radiology. 2021;299(2):278-286. Ganguli I, Keating NL, Thakore N, et al. Downstream mammary and extramammary cascade services and spending following screening breast magnetic resonance imaging vs mammography among commercially insured women. JAMA Netw Open. 2022;5(4):e227234.

Evaluator comment

Weak retrieval

Only the middle chunk carries substantive content, and even it is measured rather than a firm recommendation: it frames supplemental screening (ultrasound, MRI, DBT) for dense breasts as "an active area of research" with "limited data" in average-risk women, stopping short of endorsing MRI outright. The first and third chunks are bibliographic reference lists (citations to the DENSE trial and related MRI studies) rather than narrative guidance -- they show the evidence exists but don't state a recommendation. This looks like a chunking issue: the reference list was embedded and retrieved as if it were prose. Filtering out citation/reference-list chunks (or re-chunking around section boundaries) should improve answer quality for this query.